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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

E-CHAT: A REAL-TIME SECURE WEB-BASED CHAT APPLICATION

CSD 334 – Mini Project

Presented by: Group 5

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Problem Statement

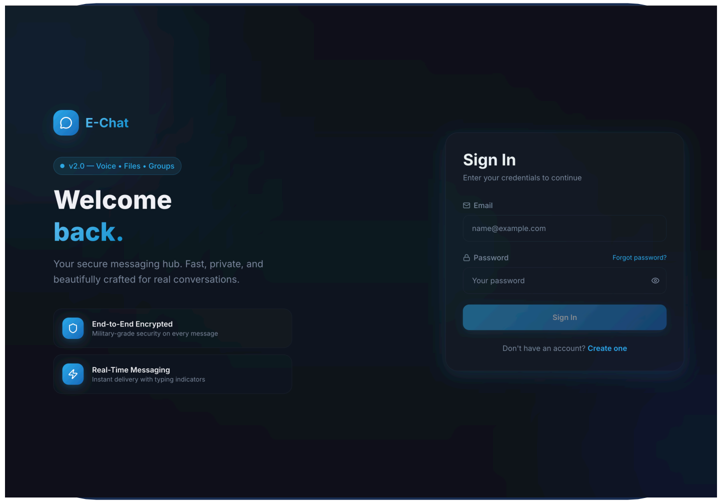
- Existing chat apps rely on centralized servers and lack true end-to-end encryption.
- Many systems struggle with scalability and real-time multimedia performance.
- Web and cross-device support is limited in current solutions.
- This project builds a secure web-based chat application for real-time communication
- It uses a TypeScript frontend, Python backend and PostgreSQL for scalability and privacy.

Aim

• To develop a secure, real-time web chat application with modern UI and scalable cloud deployment.

Objectives

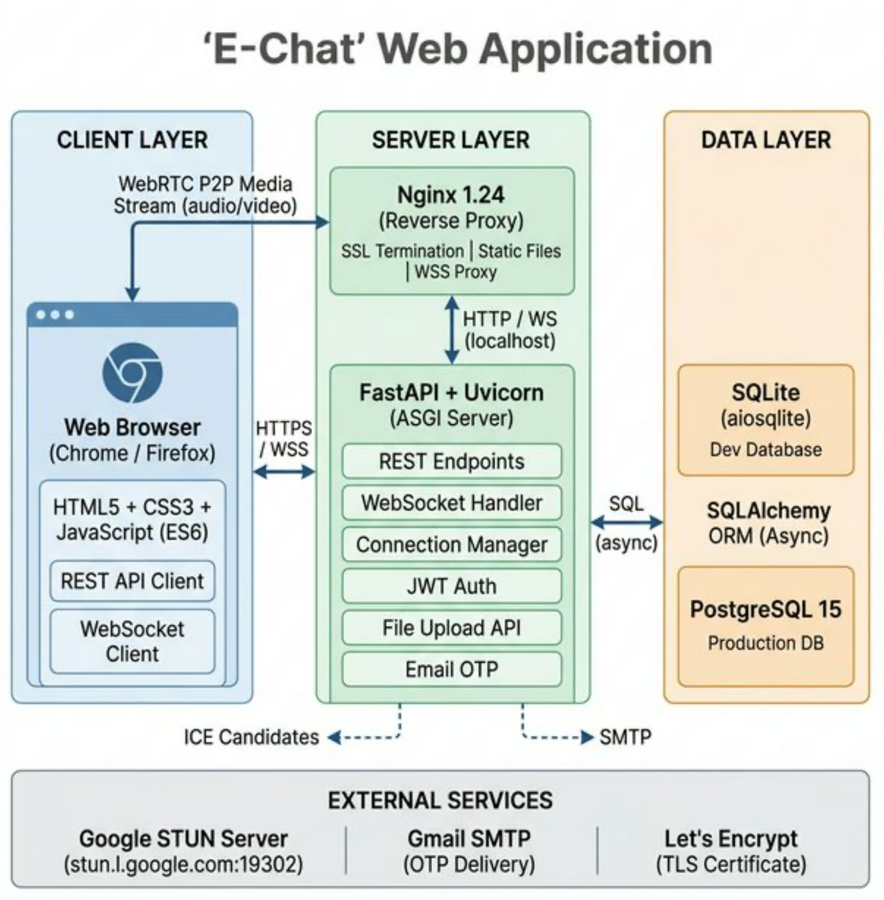
1. To implement real-time messaging using WebSocket technology
2. To provide secure user authentication using JWT tokens
3. To design a responsive and intuitive user interface
4. To ensure data persistence using a PostgreSQL database



Literature Review

S. No	Title	Authors	Journal & Year of publication	Key Points
1	Communicating and Displaying Real- Time Data with WebSocket	Pimentel V., Nickerson B.G	EEE Internet Computing (2012) vol. 16(4), pp. 45-53	• • WebSocket faster than HTTP polling for real-time data transfer with reduced latency
2	The WebSocket Protocol	Fette I., Melnikov A.	RFC 6455, IETF (2011)	• Standard protocol for bi-directional real- time communication over single TCP connection
3	JSON Web Token (JWT)	Jones M., Bradley J., Sakimura N.	RFC 7519, IETF (2015)	• Stateless authentication tokens for scalable web applications
4	How Does Your Password Measure Up? The Effect of Strength Meters on Password Creation	Ur B., Kelley P.G., Komanduri S., et al.	USENIX Security Symposium (2015) pp. 65-80	• Real-time password feedback increases strong passwords by 26%
5	Microservices Architecture: A Systematic Literature Review	Alshuqayran N., Ali N., Evans R.	. ACM Computing Surveys (2016)	• Microservices enable independent deployment and scalability for web applications

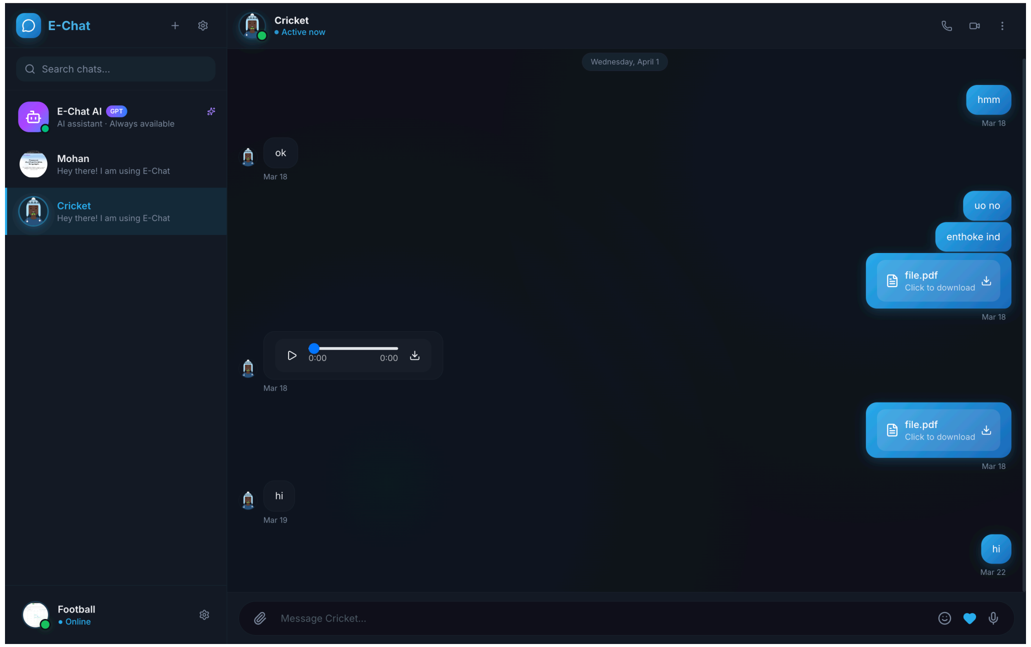
Block diagram

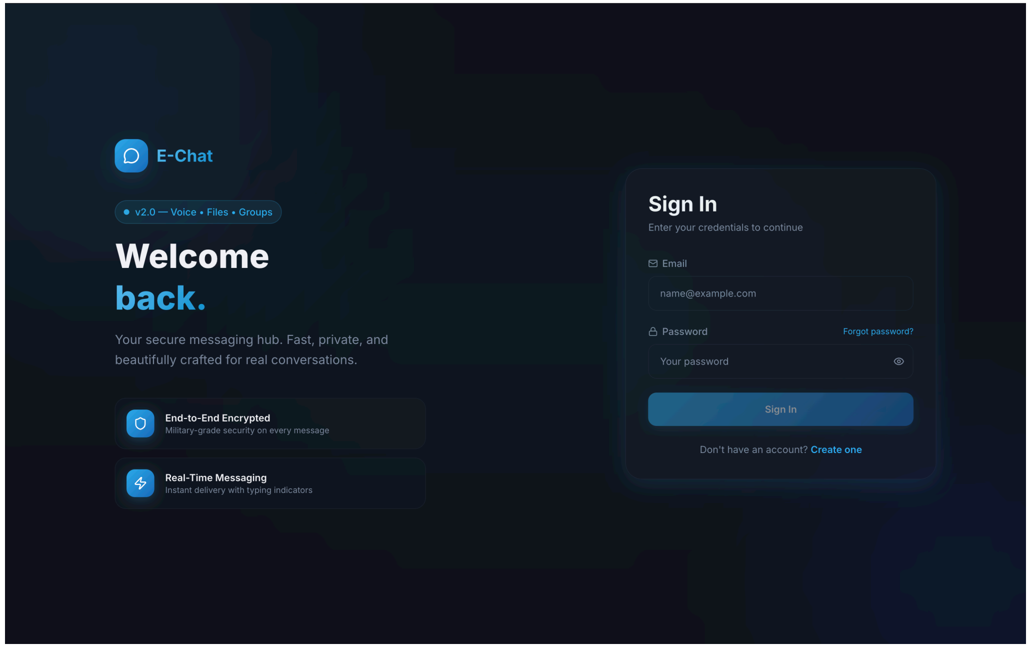


Methodology/Procedure

- **Development Approach:** Agile-based iterative model with 5 sprints over 10 weeks; continuous build, test, and improvement.
- **System Architecture:** 3-layer design – Presentation (Next.js), Communication (Socket.IO), Data (FastAPI + SQLAlchemy)
- **Real-Time Communication:** Event-driven messaging using JWT authentication with HTTP fallback
- **Audio/Video Communication:** WebRTC signaling using Offer–Answer–ICE Candidate exchange
- **Security:** JWT-based authentication with bcrypt (sha256) password hashing.
- **Database Design:** Relational model with User, Contact, Group, Message, OTP, and Call History entities
- **Communication Model:** REST APIs for authentication and data handling; Socket.IO for real-time events.
- **Frontend Strategy:** Optimistic UI rendering with server-side message confirmation.
- **Deployment:** Docker-based containerization with GitHub for version control.
- **Testing:** Functional testing and end-to-end user flow validation

Result and GUI





SDG Mapping

SDG 9	Industry, Innovation and Infrastructure	Industry, cloud systems, and IoT for better app and smarter management of food resources.
SDG 4	Quality Education	Educated the community on how to use technology.
SDG 8	Decent Work and Economic Growth	Created fair job opportunities to work on management.
SDG 11	Sustainable Cities and Communities	Supported efficient urban food management to reduce waste and build greener cities.

Roles and Responsibilities of Members

Member	Name	Role / Responsibility
Member 1	Marvin M Varghese	Project Lead & Backend Developer - FastAPI
Member 2	Muhammed Sajid	Frontend Developer & UI/UX Designer - Next.js, React, Tailwind CSS, password strength indicator, responsive design
Member 3	Parthan S	Database Administrator & Backend Support - PostgreSQL, SQL Alchemy, database schema, data modeling, query optimization
Member 4	Azeem N	DevOps Engineer & Testing Lead - Vercel/Railway deployment, CI/CD, testing, documentation, security

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Thank you